Engine Model(s): PT6A

71-00-00

POWER PLANT - CLEANING

1. General

A. Engine cleaning procedures describe the compressor desalination wash, compressor performance recovery wash, compressor turbine desalination wash, fuel nozzle cleaning, and engine external wash.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC04-001	Grease, Aircraft and Instrument
PWC05-005A	Emulsifier
PWC05-053	Plastic Sheet
PWC05-073	Water, Demineralized
PWC11-001	Cleaning Alkaline
PWC11-003	Cleaner, Detergent
PWC11-003B	Cleaner, Detergent
PWC11-014	Alcohol, Isopropyl
PWC11-027	Solvent, Petroleum
PWC11-031	Cleaner, Engine
PWC15-011	Corrosion-Inhibiting Solution
PWC15-015	Corrosion-Inhibiting Solution

3. Special Tools

The special tools listed below are used in the following procedures.

Tool No.	Name
PWC32271	Tube, Spray
PWC32677-100	Washcart, Compressor
PWC32677-300	Washcart, Compressor/fuel nozzles

4. Fixtures, Equipment and Supplier Tools

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The fixtures, equipment and supplier tools listed below are used in the following procedures.

Name
Shut off Valve
Pressure Gauge
Steel Tank
Steel Tubing
Nitrogen or Compressed Air
Pressure Gauge Adapter
Test Rig Adapter
Shut off Valve

5. Engine Washing

A. General

Depending on the operating environment and the types of deposits in the engine gas path, the following methods can be used to remove salt or dirt and other baked-on deposits that accumulate over a period of time and cause engine performance deterioration and as well as corrosion or sulphidation (Ref. 72-00-00, ENGINE - INSPECTION).

B. Types of Internal Wash

NOTE: Internal Engine washes must be done during motoring. They are done at 10 to 25% Ng and water or cleaning mixture, depending on ambient temperature (Ref. Table. 702), is injected into engine intake at a rate of 2 to 3 gal/minute (7.6 to 11.3 l/minute). Washing during a motoring cycle makes sure emulsion fluid remains in liquid form.

Compressor Desalination Wash:
This wash is used to remove salt deposits. Light dirt deposits may also be removed by this wash. The wash fluid is drinking quality water, provided minimum standards are met (Ref. Para. E.). Use either an installed compressor wash ring or a hand held wash wand to inject water into the engine intake. Ref. Para. 6. A. for the complete procedure.

Compressor Turbine Desalination Wash:
This wash is used to remove salt deposits from the compressor turbine blades and stator. The wash fluid is drinking quality water, provided minimum standards are met (Ref. Para. E.). Water is injected through a wash tube installed into the gas generator igniter boss. Ref. Para. 6. B. for the complete procedure.

NOTE: When desalination washes are done in conjunction with each other it is a must that compressor wash be done first.

Compressor Performance Recovery Wash:
This wash is used to remove more stubborn deposits which cannot be removed during normal

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desalination washes. The wash fluid includes approved chemical additives (Ref. Table. 702). This wash must only be done when engine performance loss is noticeable or trend monitoring dictates. Ref. Para. 6. C. for the complete procedure.

External Engine Wash:
This wash is used as an effective method of tracing oil leaks and as well as removing salt and other environmental contaminants from the engine external surfaces. The wash fluid is drinking quality water, provided minimum standards are met (Ref. Para. E.). Ref. Para. 7. for the complete procedure.

C. Wash Schedule Recommendations

Depending on the operating environment, the nature and frequency of engine washes are recommended in accordance with Table 701.

Table 701 Wash Schedule Recommendations

Environment	Wash	Frequency	Remarks
Continuously salt laden	Desalination (Compressor and CT) and External Engine Wash	Daily	Strongly recommended after last flight of day (See NOTES).
Occasionally operated in salt-laden or harsh environment	Desalination (Compressor and CT) and External Engine Wash	Weekly	Strongly recommended. Adjust interval to suit engine condition (See NOTES).
All	Performance Recovery	As required	Strongly recommended. Performance recovery washes are required less frequently. Adjust frequency to suit engine operating conditions as indicated by ECTM. Motoring wash for light soiling and multiple motoring washes for heavy soiling is recommended.
NOTE: 1. Multiple motoring washes should be done to the extent permitted by starter operating limitations. Observe starter cooling period (Ref. Starter Manufacturer's Manual).			
NOTE: 2. This Table covers the normal requirements of general aviation and does not			

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Environment	Wash	Frequency	Remarks
specifically consider the peculiar frequency requirements of engines operating in an agricultural environment.			
NOTE: 3. Harsh environment includes dusty, sandy or heavy polluted environments such as those near to the industrial smog or active volcanoes.			
NOTE: 4. Continually salt-laden or harsh environment also includes salt flats.			
NOTE: 5. Lubricate the linkages with light grease (PWC04-001) after engine external wash.			

D. Preparation of Solutions

Five liters (1.33 U.S. gal.) of cleaning mixture and ten liters (2.66 U.S. gal.) of rinse solution are required. The solution formulation is dependent upon prevailing ambient temperature and must be prepared in accordance with Table 702 for cleaning, and Table 703 for rinsing .

Table 702 Cleaning Solution Formulation

Ambient Temperature		Cleaning Agent (NOTE 1)	Aviation Kerosene (NOTE 2)	Isopropyl Alcohol	Water (NOTE 3)
°C	°F	% by vol.	% by vol.	% by vol.	% by vol.
Typical mixture proportions using B & B 3100 (PWC11-001C) or TC-100 (PWC11-003E), at amb. temp. of -25°C to +2°C: (Similar proportions for Magnus 1214 (PWC11-001B), R-MCS (PWC11-001D), R-MC G21 (PWC11-001E), ALMON AL-333 (PWC11-001A), TURCO 5884 (PWC11-001G) and CLIX (PWC11-001).					
+2 Up	+36 Up	25	Nil	Nil	75
-25 to +2	-12 to +36	25	15	20	40
Below -25	Below -12	25	15	40	20
Example 1. (at +2°C):					
B & B 3100 25% by vol. = 1250 ml in 5 liters (1.33 US Gallons)					
Kerosene 15% by vol. = 750 ml					
Isopropyl Alcohol 20% by vol. = 1000 ml					
Water 40% by vol. = 2000 ml					
Total 100% by vol. = 5000 ml or 5 liters (1.33 US Gallons)					
Typical mixture proportions using Turco 4217 (PWC11-001F) or Ardrox 624 (PWC11-001H).					

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Ambient Temperature		**Cleaning Agent (NOTE 1)**	**Aviation Kerosene (NOTE 2)**	**Isopropyl Alcohol**	**Water (NOTE 3)**
°C	**°F**	**% by vol.**	**% by vol.**	**% by vol.**	**% by vol.**
+2 Up	+36 Up	4	40	Nil	56
-25 to +2	-12 to +36	4	40	20	36
Below -25	Below -12	4	40	36	20
Example 2. (at +2°C):					
Turco 4217 4% by vol. = 200 ml in 5 liters (1.33 US Gallons)					
Kerosene 40% by vol. = 2000 ml					
Isopropyl Alcohol 0% by vol. = 0 ml					
Water 56% by vol. = 2800 ml					
Total 100% by vol. = 5000 ml or 5 liters (1.33 US Gallons)					
Typical mixture proportions using Ardrox 6345 (PWC11-003) or Ardrox 6367 (PWC11-003B), Turboclean 2 (PWC11-003C) and ZOK 27 (PWC11-003D)					
+5 Up	+41 Up	20	Nil	Nil	80
-5 to +5	+23 to +41	20	Nil	20	60
-21 to -5	-6 to +23	20	Nil	30	50
Below -21	Below -6	20	Nil	40	40
Example 3. (at +5°C):					
Zok 27 20% by vol. = 1000 ml in 5 liters (1.33 US Gallons)					
Isopropyl Alcohol 0% by vol. = 0 ml					
Water 80% by vol. = 4000 ml					
Total 100% by vol. = 5000 ml or 5 liters (1.33 US Gallons)					
NOTE: 1. Use any cleaning agent listed (Ref. CONSUMABLE MATERIALS). Solution strength must be in accordance with manufacturer/supplier's recommendations. Where no specific instructions are provided, proportion of cleaning agent should be as indicated.					
NOTE: 2. Use of emulsifying agent (PWC05-005A), 3% by vol. or 150 ml in 5 liters is					

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Ambient Temperature		Cleaning Agent (NOTE 1)	Aviation Kerosene (NOTE 2)	Isopropyl Alcohol	Water (NOTE 3)
°C	°F	% by vol.	% by vol.	% by vol.	% by vol.
highly recommended to prevent possibility of kerosene separation.					
NOTE: 3. Drinking quality water is permissible for motoring wash procedures.					

Table 703 Rinse Solution Formulation

Ambient Temperature		Water (See NOTE)	Isopropyl Alcohol
°C	°F	% by vol.	% by vol.
+2 up	+36 up	100	-
-25 to +2	-12 to +36	50	50
Below -25	Below -12	40	60
NOTE: 1. Drinking quality water is permitted. Refer to Equipment Required for acceptance criteria.			

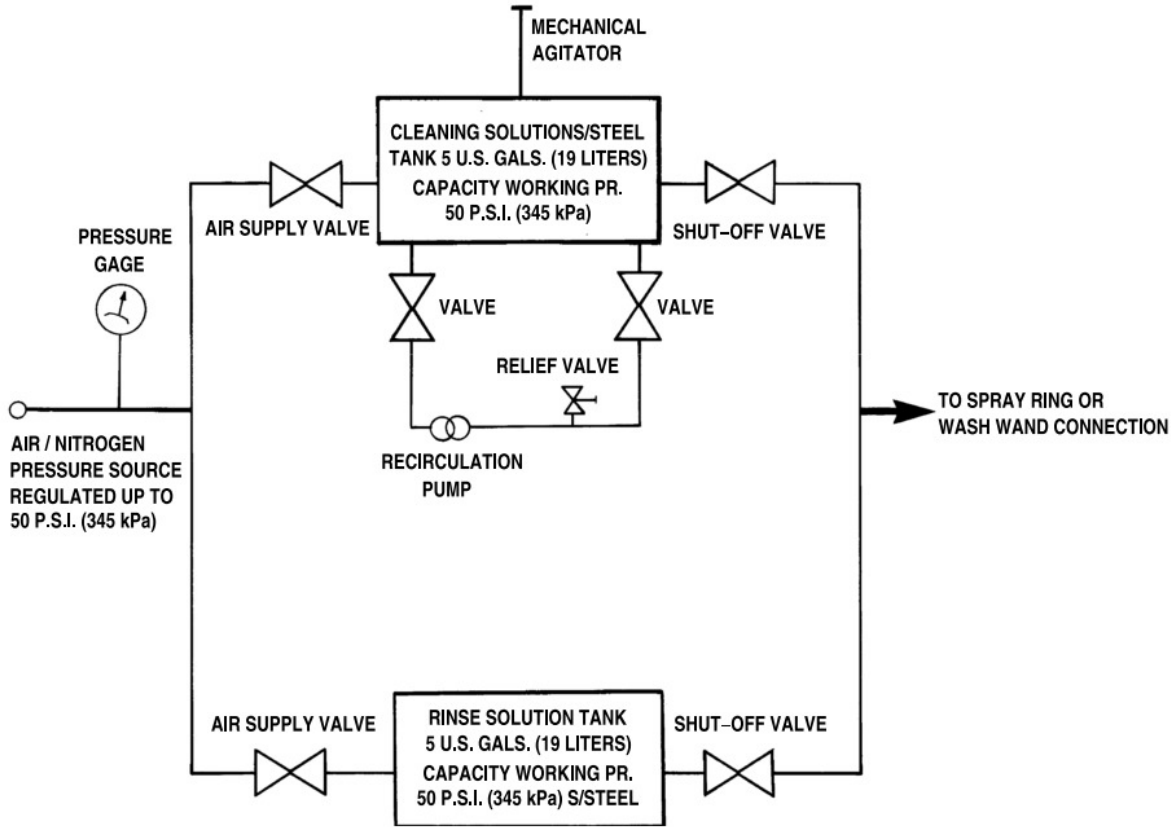
E. Equipment Required (Ref. Figs. 701, 702 and 703)

CAUTION: DO NOT USE ANY RESTRICTOR WHEN THE MOTORING WASH IS DONE.

NOTE: Some engines have a built-in, 50-hole, deluge washing ring as standard installation feature.

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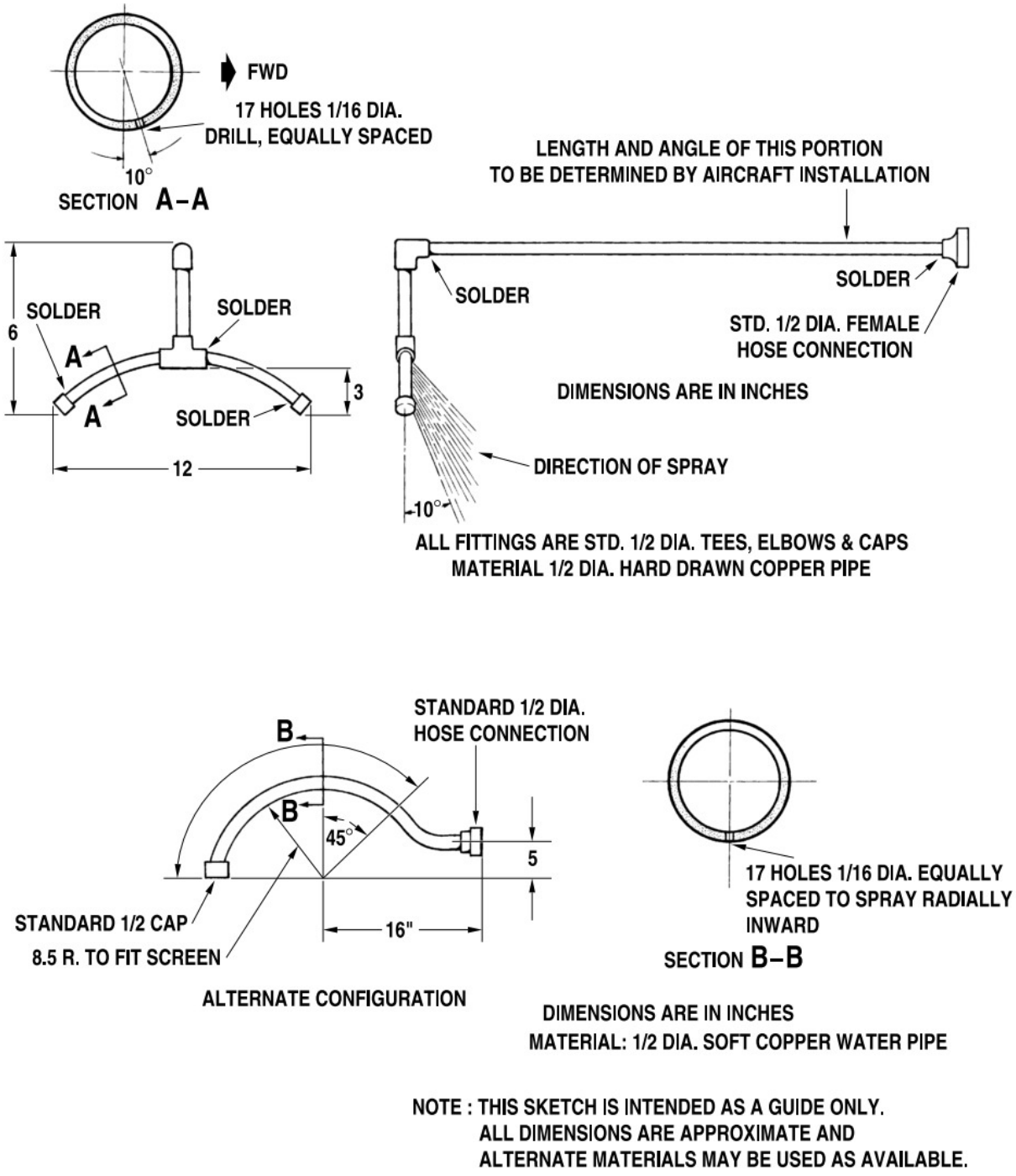
Figure 701 Compressor Wash Rig - Schematic



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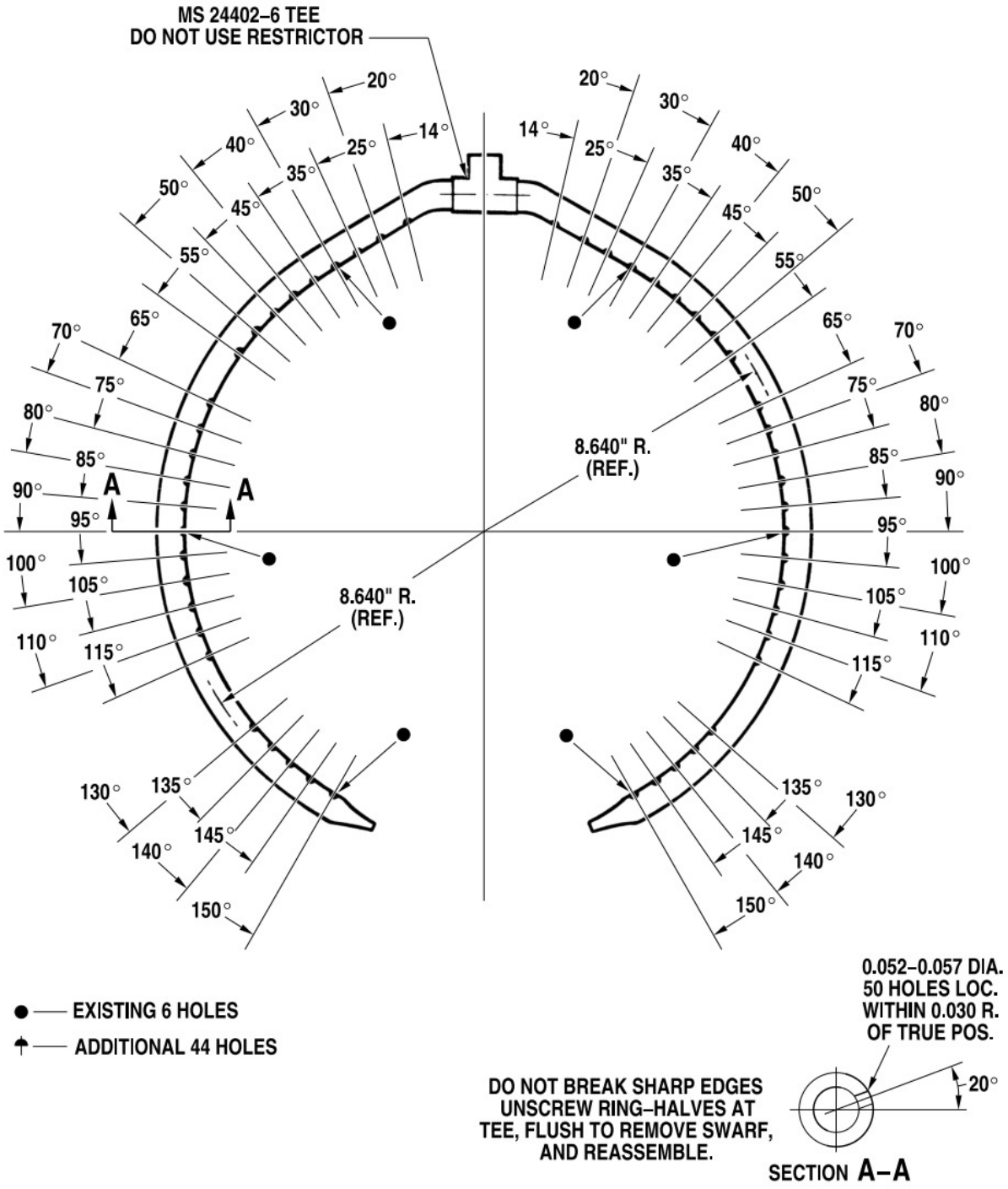
Figure 702 Water Wash Wand



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Figure 703 Modification of Standard - 6-Hole Wash Ring to 50-Hole



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(1) Wash wand (Ref. Fig. 702) or wash ring (Ref. Fig. 703). Flow rate 2 to 3 gpm (U.S.) (7.57 to 11.33 l/min).

(2) Modified 50-hole wash ring (Ref. Fig. 703) (if not original equipment). Flow rate 2 to 3 gpm (U.S.) (7.57 to 11.33 l/min).

(3) Wash cart (PWC32677-100) or (PWC32677-300), optional locally manufactured wash rig (Ref. Fig. 701) comprising the following:

(a) Two stainless steel tanks, 5 gallon (U.S.) (19 liters) capacity, each capable of withstanding up to 50 psig (345 kPa) working pressure.

(b) Mechanical agitator, or recirculation pump with relief valve, for mixture agitation.

(c) Two flow control valves, for use with recirculation pump.

(d) Two air supply shut-off valves.

(e) Two fluid delivery shut-off valves.

(f) One pressure gage located in air delivery line.

(g) Suitable tubing to interconnect components.

NOTE: Valve and tubing connections are to be 5/16-inch (8mm) ID minimum (Ref. TOOLS/FIX/EQUIP).

(4) Compressed air or nitrogen supply, regulated up to 50 psig (345 kPa).

(5) A source of demineralized water (PWC05-073) to the following specifications:

(a) Appearance: free of suspended solids.

(b) Total solids: 10 ppm (mg/l) Max.

(c) Specific conductance: 11 micro-mho/cm Max.

(d) Silica content: 3 ppm (mg/l) Max.

(e) pH value: 5.0 to 7.5 inclusive.

(f) Intake filter not coarser than 10 microns.

(6) Drinking quality water criteria:

(a) Appearance: free of visible extraneous impurities.

(b) Total solids: 175 ppm (mg/l) Max.

(c) pH value: 6.0 to 8.0 inclusive.

(d) Chlorides: 15 ppm (mg/l) Max.

(e) Sulfates: 10 ppm (mg/l) Max.

NOTE: Drinking water quality varies according to location and season; these specifications are provided as a guide only.

6. Engine Motoring Washes

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A. Compressor Desalination Wash

NOTE: P&WC recommends to do a compressor turbine desalination wash after the compressor desalination wash for salt-laden or harsh environment, since the compressor desalination wash will transfer salt deposits into the turbine (Ref. SIL No. PT6A - 206).

(1) Depending on the ambient temperature, fill wash tanks with the appropriate solution (Ref. Table 702). Alternatively, connect suitable hose to drinking water tap for ambient temperature above +2°C (+36°F).

(2) Connect compressed air regulated to 30 to 50 psig (207 to 345 kPa) to wash tanks. If tap water is used, connect through centrifugal pump if water pressure is below 30 psig (207 kPa).

(3) Remove or open engine cowlings, as applicable, to expose the engine inlet screen and install the wash wand or modified wash ring (Ref. Figs. 702 and 703).

(4) Connect the wash ring or wand to the pressurized tanks, or to the drinking water tap as applicable.

NOTE: To prevent the precipitation of deposits due to the use of "hard" water, allow the engine to cool below 65°C (150°F). The minimum duration for engine cooling is 40 minutes since the last operation.

CAUTION: CONTAMINATION/BLOCKAGE OF THE FCU PNEUMATIC SECTION, OR THE P3 FILTER WILL AFFECT ENGINE HANDLING. REMOVE THE AIR TUBE FROM THE ENGINE BEFORE PERFORMING A MOTORING WASH IF SB1495 IS NOT INSTALLED.

CAUTION: FOR Pre-SB1495 ENGINES: SPECIAL CARE SHOULD BE TAKEN WHEN A COMPRESSOR WASH IS DONE TO MAKE SURE THAT THE P3 LINE IS COMPLETELY REMOVED TO AVOID BENDING STRESS OF ONE COUPLING.

(5) For Pre-SB1495 Engines: Remove forward P3 air tube (Ref. 73-10-07) from engine to prevent distortion of tube.

NOTE: For Post-SB1495 Engines: Removal of the forward P3 air tube is not required as the drain valve installed in the P3 filter cover will drain remaining fluids.

(6) Make sure the ignition switch and aircraft bleed air are OFF.

CAUTION: DO NOT MOTOR FOR MORE THAN 30 SECONDS; OBSERVE THE STARTER COOLING PERIOD (REF. STARTER MANUFACTURERS MANUAL). WHEN USING A WATER/ALCOHOL SOLUTION, PERFORM AN ADDITIONAL 30-SECOND DRY MOTORING RUN BETWEEN STEPS (10) AND (11) FOLLOWING, TO PURGE THE ENGINE OF VOLATILE FUMES. MAKE SURE THE PRESCRIBED STARTER COOLING PERIOD IS ALLOWED FOLLOWING THIS ADDITIONAL STEP.

(7) Motor engine (starter only).

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CAUTION: DO NOT SPRAY WASH MIXTURE OR WATER INTO THE ENGINE INTAKE AT A RATE MORE THAN THREE GALLON/MINUTE (11.3 LITER/MINUTE) TO PREVENT FCU CONTAMINATION.

(8) When the Ng reaches 5% minimum, inject the wash mixture or water into the engine at a rate of two to three gallon/minute (7.6 - 11.3 liter/minute).

(9) Stop motoring after 30 seconds.

(10) Shut off the cleaning solution supply as soon as the Ng falls to 5%.

(11) Observe the starter cooling period (Ref. Starter Manufacturers Manual).

(12) Remove the wash wand or ring from the inlet screen. Reinstall or close the engine cowlings.

(13) For Pre-SB1495 Engines: Install the front air tube (Ref. 73-10-07).

NOTE: It is important to make sure all solid P3 lines are properly and correctly fitted before the coupling nuts are tightened. The coupling nuts should be seated without the use of tools (finger tight) before torquing. The installation of any pipe brackets should not distort the pipe in any way.

(14) Start the engine (Ref. ADJUSTMENT/TEST). Run the engine at 80% Ng for one minute or more to completely dry the engine. Check engine acceleration (Ref. ADJUSTMENT/TEST).

(15) Shut down the engine (Ref. ADJUSTMENT/TEST).

B. Compressor Turbine Desalination Wash

CAUTION: IF TURBINE WASH IS TO BE DONE IN CONJUNCTION WITH COMPRESSOR WASH, WASH THE COMPRESSOR FIRST.

(1) Depending on the ambient temperature, fill the wash tanks with the appropriate cleaning solution (Ref. Table 702).

(2) Connect the compressed air supply or nitrogen supply, regulated 30 to 50 psig (207 to 345 kPa) to wash tanks.

(3) Remove or open the engine cowlings, as applicable (Ref. Aircraft Maintenance Manual), to expose engine exciter box and igniters.

CAUTION: RESIDUAL VOLTAGE IN IGNITION EXCITER MAY BE DANGEROUSLY HIGH. MAKE SURE THAT IGNITION HAS BEEN SWITCHED OFF FOR A MINIMUM OF 6 MINUTES. ALWAYS DISCONNECT COUPLING NUTS AT IGNITION EXCITER FIRST AND USE INSULATED TOOLS TO REMOVE SAME. DO NOT TOUCH OUTPUT CONNECTORS OR COUPLING NUTS WITH BARE HANDS.

(4) Remove most accessible igniter and gasket. Discard gasket.

CAUTION: SUPPORT DELIVERY HOSE TO PREVENT DAMAGE TO WASH TUBE.

(5) Install turbine wash tube (PWC32271) in igniter boss, with RGB arrow inscribed on tang pointing toward reduction gearbox. Tighten fingertight.

(6) With tube connection installed, connect to pressurized tank or drinking water supply.

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Delivery hose must be supported to avoid damage to wash tube.

NOTE: Engine temperature must be below 65 °C (150 °F) to make sure inadvertent use of hard water does not result in precipitation of deposits. Minimum cooling off period is 40 minutes since engine last operated.

CAUTION: CONTAMINATION/BLOCKAGE OF THE FCU PNEUMATIC SECTION, OR THE P3 FILTER WILL AFFECT ENGINE HANDLING. REMOVE THE AIR TUBE FROM THE ENGINE BEFORE PERFORMING A MOTORING WASH IF SB1495 IS NOT INSTALLED.

CAUTION: FOR Pre-SB1495 ENGINES: SPECIAL CARE SHOULD BE TAKEN WHEN A COMPRESSOR WASH IS DONE TO MAKE SURE THAT THE P3 LINE IS COMPLETELY REMOVED TO AVOID BENDING STRESS OF ONE COUPLING.

(7) For Pre-SB1495 Engines: Remove forward P3 air tube (Ref. 73-10-07) from engine to prevent distortion of tube.

NOTE: For Post-SB1495 Engines: Removal of the forward P3 air tube is not required as the drain valve installed in the P3 filter cover will drain remaining fluids.

(8) Make sure the ignition and aircraft bleed air are OFF.

(9) Make sure propeller is not tethered and is free to turn.

CAUTION: DO NOT MOTOR FOR MORE THAN 30 SECONDS. OBSERVE STARTER COOLING PERIOD (REF. STARTER MANUFACTURERS MANUAL).

(10) Motor the engine (starter only).

(11) When the Ng reaches 5%, inject the cleaning solution into the engine.

(12) Stop motoring after 30 seconds.

(13) Shut off the cleaning solution supply as soon as the Ng falls to 5%.

(14) If water/alcohol solution is used, observe starter cooling period and then motor the engine for 30 seconds.

(15) Disconnect and remove the cleaning solution equipment, and remove wash tube from igniter boss.

(16) Install igniter with new gasket (Ref. 74-20-04).

(17) For Pre-SB1495 Engines: Install the front air tube (Ref. 73-10-07).

NOTE: It is important to make sure all solid P3 lines are properly and correctly fitted before the coupling nuts are tightened. The coupling nuts should be seated without the use of tools (finger tight) before torquing. The installation of any pipe brackets should not distort the pipe in any way.

(18) Check for zero leakage during engine drying cycle. If satisfactory, lockwire all coupling nut (Ref. Step (16)).

C. Compressor Performance Recovery Wash (Ref. Fig. 701)

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NOTE: P&WC recommends to do a compressor turbine desalination wash after the compressor performance recovery wash for salt laden environment, since the compressor performance recovery wash will transfer salt deposits into the turbine (Ref. SIL No. PT6A - 206).

(1) Depending on the ambient temperature, fill the wash tanks with the appropriate cleaning solution (Ref. Table 702) and rinse solution (Ref. Table 703) in separate tanks.

(2) Connect the compressed air supply, 30 to 50 psig (207 to 345 kPa), to wash tanks.

(3) Remove or open the engine cowlings, as applicable, to expose the engine inlet screen and install the wash ring or wand.

(4) Connect the wash ring or wand, as applicable, to the pressure tanks.

NOTE: To prevent the precipitation of deposits due to the use of "hard" water, allow the engine to cool below 65 °C (150 °F). The minimum duration for engine cooling is 40 minutes since the last operation.

CAUTION: CONTAMINATION/BLOCKAGE OF THE FCU PNEUMATIC SECTION, OR THE P3 FILTER WILL AFFECT ENGINE HANDLING. REMOVE THE AIR TUBE FROM THE ENGINE BEFORE PERFORMING A MOTORING WASH IF SB1495 IS NOT INSTALLED.

CAUTION: FOR Pre-SB1495 ENGINES: SPECIAL CARE SHOULD BE TAKEN WHEN A COMPRESSOR WASH IS DONE TO MAKE SURE THAT THE P3 LINE IS COMPLETELY REMOVED TO AVOID BENDING STRESS OF ONE COUPLING.

(5) For Pre-SB1495 Engines: Remove forward P3 air tube (Ref. 73-10-07) from engine to prevent distortion of tube.

NOTE: For Post-SB1495 Engines: Removal of the forward P3 air tube is not required as the drain valve installed in the P3 filter cover will drain remaining fluids.

(6) Make sure the ignition and aircraft bleed air are OFF.

CAUTION: DO NOT MOTOR FOR MORE THAN 30 SECONDS; OBSERVE THE STARTER COOLING PERIOD (REF. STARTER MANUFACTURERS MANUAL). WHEN USING A WATER/ALCOHOL SOLUTION, PERFORM AN ADDITIONAL 30-SECOND DRY MOTORING RUN BETWEEN STEPS (13) AND (14) FOLLOWING, TO PURGE THE ENGINE OF VOLATILE FUMES. MAKE SURE THE PRESCRIBED STARTER COOLING PERIOD IS ALLOWED FOLLOWING THIS ADDITIONAL STEP.

(7) Motor the engine (starter only).

CAUTION: DO NOT SPRAY CLEANING SOLUTION INTO THE ENGINE INTAKE AT A RATE MORE THAN THREE GALLON/MINUTE (11.3 LITER/MINUTE) TO PREVENT FCU CONTAMINATION.

(8) When the Ng reaches 5%, inject the cleaning solution into the engine at a rate of two to three gallon/minute (7.6 - 11.3 liter/minute).

(9) Stop motoring after 30 seconds.

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(10) Shut off the cleaning solution supply as soon as the Ng falls to 5%.

(11) Allow the cleaning solution to soak for 15 to 30 minutes.

CAUTION: DO NOT SPRAY RINSE SOLUTION INTO THE ENGINE INTAKE AT A RATE MORE THAN THREE GALLON/MINUTE (11.3 LITER/MINUTE) TO PREVENT FCU CONTAMINATION.

(12) Motor the engine. When the Ng reaches 5%, inject the rinse solution into the engine (half quantity only) at a rate of two to three gallon/minute (7.6 - 11.3 liter/minute).

(13) Observe the starter cooling period. (Ref. Starter Manufacturers Manual).

(14) Repeat step (12) observing the starter limitations.

(15) Remove the wash ring or wand from the inlet screen.

(16) For Pre-SB1495 Engines: Install the front air tube (Ref. 73-10-07).

NOTE: It is important to make sure all solid P3 lines are properly and correctly fitted before the coupling nuts are tightened. The coupling nuts should be seated without the use of tools (finger tight) before torquing. The installation of any pipe brackets should not distort the pipe in any way.

(17) Switch the ignition ON and start the engine (Ref. ADJUSTMENT/TEST). Run the engine at 80% Ng for one minute or more to completely dry the engine. Check for zero air leakage during the drying cycle. Check engine acceleration.

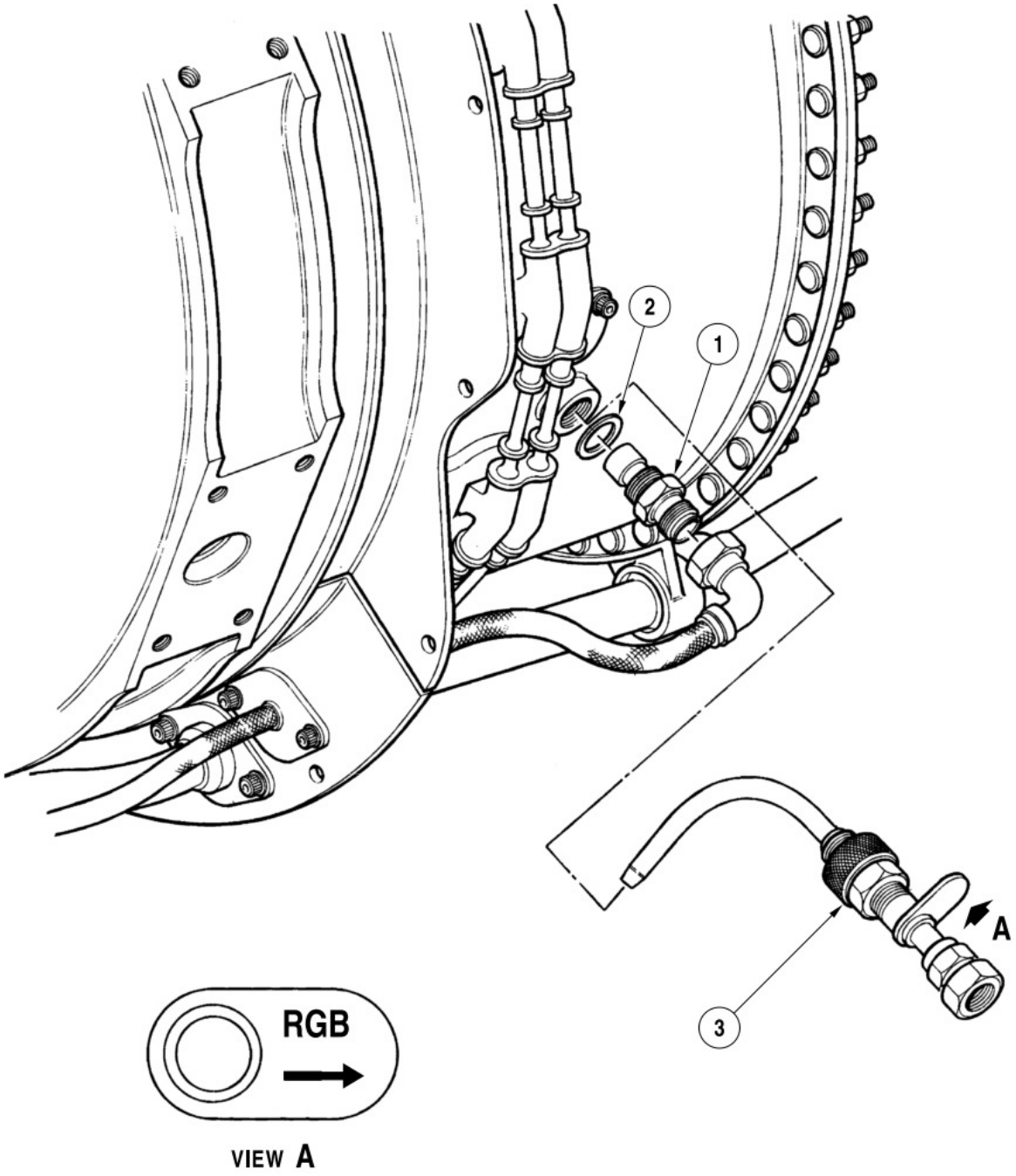
(18) Shut down the engine (Ref. ADJUSTMENT/TEST).

(19) Install or close the cowling (Ref. Aircraft Maintenance Manual).

7. Engine External Wash

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Figure 704 Removal/Installation Compressor Turbine Wash Tube Connection (Typical)



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1. Igniter/Glow Plug
2. Gasket
3. Tube Sprayer (PWC32271)

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A. General

CAUTION: DO NOT USE GASOLINE OR SIMILAR TOXIC SUBSTANCES FOR EXTERNAL ENGINE CLEANING.

CAUTION: DO NOT ATTEMPT TO WASH AN ENGINE THAT IS STILL HOT OR RUNNING. LET ENGINE COOL FOR A MINIMUM OF 40 MINUTES AFTER SHUTDOWN.

Fresh water external washing is very effective in tracing possible origin of external oil leakage, and is recommended when the engine is contaminated with salt or corrosive chemicals such as those found in industrial smog. Demineralized water is not necessary. The engine should never be left in a contaminated condition overnight, or longer.

For engine operations in corrosive environments it is recommended to perform an external engine wash per the schedule in Table .701 .

NOTE: When you operate the engine in salt-laden or harsh environment, apply corrosion inhibitor on the external components for more protection (Ref. 72-00-00, ENGINE - INSPECTION).

If water wash is ineffective, a petroleum solvent (varsol/water emulsion), (PWC11-027) or (PWC11-031), may be used to remove oil and grime. Thoroughly rinse with water to remove all traces of cleaning fluid. Completely dry the engine using clean, dry compressed air.

8. Compressor Turbine Wash

A. General

This is a method for washing compressor turbine blades, while installed in the engine, to alleviate sulphidation attack and salt deposits.

Water or water/alcohol, depending upon ambient temperatures (Ref. Table 702), is used to wash the compressor turbines.

Depending on the operating environment, the nature and frequency of wash carried out are recommended to be in accordance with Table 701 .

Approximately 1.90 liters (0.5 U.S. gallon) will pass through the compressor turbine during a 30-second motoring wash cycle.

B. Equipment

For equipment required for a motoring wash, reference Para. E..

The wash tube assembly (PWC32271) is designed to introduce the wash solution directly to the compressor turbine blades.

C. Compressor Turbine Wash Method

(1) Depending on the ambient temperature, fill the wash tank with the appropriate mixture (Ref. Table 702). Alternatively, at temperatures above 2°C (36°F) use a suitable hose connected to a drinking water supply.

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(2) Connect the compressed air or nitrogen supply, regulated to 30 to 50 psig (207 to 345 kPa) to the wash tanks. If drinking water is used, connect through a centrifugal pump if water pressure is below 30 psig (207 kPa).

WARNING: RESIDUAL VOLTAGE IN IGNITION EXCITER MAY BE DANGEROUSLY HIGH. MAKE SURE THAT THE IGNITION IS SWITCHED OFF. ALWAYS DISCONNECT THE COUPLING NUTS AT THE IGNITION EXCITER FIRST. ALWAYS USE INSULATED TOOLS TO REMOVE THE COUPLING NUTS. DO NOT TOUCH THE OUTPUT CONNECTORS OR COUPLING NUTS WITH BARE HANDS.

(3) Remove the most accessible igniter (1, Fig. 704) and the gasket (2) (Ref. 74-20-00, IGNITION CABLES AND SPARK IGNITERS - MAINTENANCE PRACTICES).

(4) Install the compressor turbine wash tube assembly (3) fingertight in the igniter port. Make sure the arrow and "RGB" on the tang point toward the reduction gearbox and the tang is parallel with the engine centerline.

CAUTION: SUPPORT THE DELIVERY HOSE TO PREVENT DAMAGE TO THE SPRAY TUBE.

(5) Connect the pressurized tank or drinking water supply to the wash tube assembly.

NOTE: A minimum cool-down period of 40 minutes should be observed after the engine run. Engine temperature must be below 65 °C (150 °F) to make sure that inadvertent use of hard water does not result in the precipitation of deposits.

(6) Make sure the ignition and aircraft bleed air are OFF.

CAUTION: DO NOT MOTOR THE ENGINE FOR MORE THAN 30 SECONDS; OBSERVE THE STARTER COOLING PERIOD (REF. STARTER MANUFACTURER'S MANUAL).

(7) Do a dry motoring run (Ref. ADJUSTMENT/TEST).

CAUTION: WHEN USING A WATER/ALCOHOL SOLUTION, DO AN ADDITIONAL DRY MOTORING RUN TO PURGE THE ENGINE. OBSERVE THE STARTER LIMITS.

(8) When the Ng reaches 5%, inject the water or water/alcohol as applicable, into the engine.

(9) Stop the motoring run after 30 seconds.

(10) Shut off the water or water/alcohol as the Ng decreases through 5%.

CAUTION: OBSERVE THE STARTER COOLING PERIOD (REF. STARTER MANUFACTURER'S MANUAL).

(11) Repeat the washing cycles as necessary to remove contaminants from the turbines.

(12) Disconnect the pressurized tanks or drinking water supply from the wash tube assembly (3, Fig. 704) and remove the assembly from the igniter port.

(13) Install the igniter (1) with the gasket (2) into the igniter port (Ref. 74-20-00, IGNITION CABLES AND SPARK IGNITERS - MAINTENANCE PRACTICES).

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CAUTION: OBSERVE THE STARTER COOLING PERIOD (REF. STARTER MANUFACTURER'S MANUAL).

(14) Do an additional 30 second dry motoring run if the water/alcohol has been used.

(15) Start the engine (Ref. ADJUSTMENT/TEST) and run at 80% Ng for one minute or more to completely dry the engine.

(16) Shut down the engine.

9. Fuel Nozzle In-situ Cleaning

A. General

Wash cart (PWC32677-300) is required. Prior to the wash cycle, the cleaning rig must be prepared and operated in accordance with the Cleaning Rig Manual.

Cleaning should be initiated on new or recently cleaned nozzles as the procedure will not clear previous blockages. Clean every 200 hours. Adjust the interval based on condition and experience.

B. Preparation

CAUTION: CLEANING AGENT IS AN ALKALINE SUBSTANCE THAT CAN CAUSE SKIN IRRITATION. AVOID SKIN CONTACT.

(1) Mixture ratios:

(a) At ambient temperatures above 2°C (36°F) mix by volume: 1 part cleaner (PWC11-001G) to 4 parts demineralized water (PWC05-073).

(b) At ambient temperatures below 2°C (36°F) mix a 50% solution of alcohol(PWC11-014) and demineralized water (PWC05-073), then mix 4 parts of water/alcohol (by volume) to 1 part cleaner (PWC11-001G).

CAUTION: MAKE SURE THAT AIR HAS BEEN RELEASED FROM RESERVOIRS.

(2) Fill reservoirs with an appropriate volume of cleaning solution.

(3) Fill rinse reservoir with appropriate volume of demineralized water or demineralized water/alcohol mixture according to ambient temperature.

(4) Remove two gas generator drain valves from engine (Ref. 73-10-06). Install suitable drain hoses.

(5) Disconnect fuel inlet line from flow divider and purge valve outlet elbow (Ref. 73-10-04).

(6) Connect wash rig fluid delivery hose to valve outlet elbow using adapter (0.4375-20 UNJF).

(7) Connect rig fluid delivery hose to rig "FUEL NOZZLE" quick disconnect.

(8) Connect compressed air.

CAUTION: DO NOT EXCEED 120 PSIG.

(9) Set wash rig to 60 psig using reservoir regulator, and pulse to 70 psig using pulse regulator.

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C. Cleaning

(1) Let the engine cool for a minimum of 40 minutes after engine shutdown before cleaning the fuel nozzles.

(2) Place valve A and B to AGITATE and valve C OFF, and agitate detergent solution for two minutes.

(3) Place valve A and B to WASH and valve C ON, and detergent wash for three minutes.

(4) Place all valves OFF and let soak for ten minutes.

(5) Place valve A and B to WASH and valve C ON, and detergent wash for three minutes.

(6) Place all valves OFF and let soak for ten minutes.

(7) Place valve A and B to RINSE and valve C ON, and rinse for five minutes.

(8) Place valve A and B OFF and valve C to AIR DRY, and dry for five minutes.

(9) Place valves OFF and reduce both reservoir and pulse pressure to zero.

(10) Remove compressed air supply from wash rig.

(11) Remove wash rig fluid delivery hose from flow divider elbow.

(12) Reconnect dump line or air inlet line (Ref. 73-10-04).

(13) Remove blanking cap at fuel inlet line and elbow on flow divider and dump or purge valve.

(14) Reconnect fuel inlet line (Ref. 73-10-04).

(15) Remove drain hoses and reinstall gas generator case drain valves (Ref. 73-10-06).

(16) Do an engine ground run and check for leaks (Ref. ADJUSTMENT/TEST).

(17) Clean wash rig, stow fluid delivery hoses, and bleed the air pressure from the reservoirs.

10. Inhibition of Engine Corrosion

A. Optional Application of Engine Corrosion-inhibiting Solution

(1) Information relocated to Chapter 72-00-00, Inspection.

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